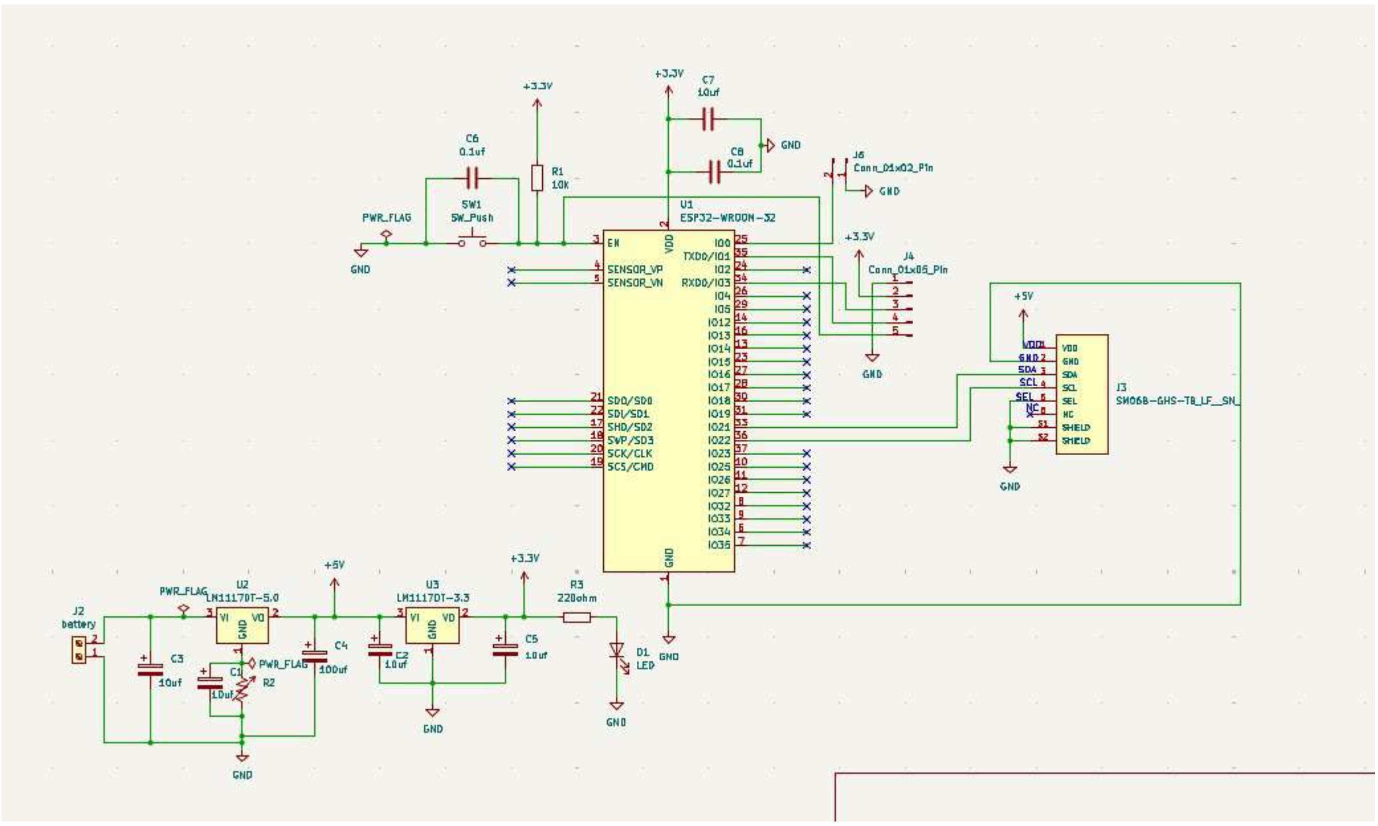
Part	Connector
Sensor side	ACES 51451-0060N-001
Cable side	Compatible with ACES 51452-006H0H0-001 (e.g., JST GHR-06V-S)

Table 10: SEN5x connector

Pin	Name	Description	Comments
1	VDD	Supply voltage	5V ± 10%
2	GND	Ground	
3	SDA	Serial data input / output	LVTTL 3.3V compatible
4	SCL	Serial clock input	LVTTL 3.3V compatible
5	SEL	Interface select	Connect to GND
6	NC	Do not connect	



+3.3V
C7
10uf
C6
0.1uf
C8
0.1uf
GND
J6
Conn_01x02_Pin
R1
10k
SW1
SW_Push
PWR_FLAG
U1
ESP32-WROOM-32
EN
VDD
SENSOR_VP
SENSOR_VN
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
J4
Conn_01x05_Pin
+5V
J3
SM06B-GHS-TB_LF__SN_
VDD
SDA
SCL
SEL
NC
SHIELD
J2
battery
U2
LM1117DT-5.0
VI
VO
C3
10uf
C1
10uf
R2
C4
100uf
U3
LM1117DT-3.3
C2
10uf
C5
10uf
R3
220ohm
D1
LED

Synopsis

"A compact PCB based solution for accurately sensing and monitoring air quality in real time."

Group Members:

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Ashutosh Swamy
Naman Vangani
Shlok Parge

Under the Guidance of

Prof. Manisha Kuveskar

The entire project has been **sponsored** by **ChipIOT Embedded Solutions**

In partial fulfillment of

Int. B.Tech. ECE (AI&ML)
T.Y. Diploma

ABSTRACT

Air quality is crucial for public health and environmental balance, yet pollution from industrialization, urban growth, and vehicles continues to rise, causing respiratory problems, cardiovascular diseases, allergies, and contributing to issues like smog and climate change. Most people remain unaware of daily pollution levels due to the lack of affordable personal monitoring tools. This project presents a compact, low-cost air quality monitoring device that uses advanced sensors to measure PM2.5, PM10, CO, CO_2, temperature, and humidity, providing real-time, accurate data through a simple interface. The system helps users take timely actions such as improving ventilation or reducing outdoor exposure and is suitable for homes, schools, offices, hospitals, and public spaces. By making monitoring accessible, it promotes awareness, healthier living, and supports broader environmental conservation efforts.

KEYWORDS

Air Quality Index (AQI), ESP32, SEN55, IoT System, PM0.5, PM1.0, PM2.5, PM4.0, PM10, Volatile Organic Compounds (VOCs).

INTRODUCTION

Air quality has become a growing concern due to rapid urbanization, industrial activities, and rising pollution levels that directly affect human health and environmental stability. Continuous and accurate monitoring is essential to understand changing environmental conditions, detect harmful pollutants, and implement timely measures to safeguard public well-being. This project aims to design a portable, intelligent, and user-friendly system capable of detecting and evaluating various air pollutants in real time with high precision. By combining modern sensing technologies with IoT-based connectivity, the system not only measures pollutant levels but also stores historical data, allowing users to observe long-term trends, analyze patterns, and assess the impact of different environmental factors. Through cloud integration and mobile accessibility, the system enables users to remotely monitor air quality, identify pollution sources, and receive alerts during high-risk conditions. Overall, the project empowers individuals, communities, and organizations to make informed decisions, improve their surroundings, and contribute to creating healthier and more sustainable living environments.

LITERATURE SURVEY

Zhang and Woo presented an IoT-based air quality monitoring and prediction system using both mobile and fixed sensors to improve spatial coverage. Their approach provided accurate forecasting but required constant cloud connectivity and high power, making it less suitable for small portable devices. **[1]**

Zheng et al. developed a low-power wide-area (LPWA) air quality monitoring system that achieved long-range communication. However, the system suffered from higher latency and slower updates, which reduced its usefulness for detecting quick changes in nearby air quality. **[2]**

Chen et al. proposed a participatory PM2.5 monitoring framework that relied on citizen-operated sensors. While it increased data volume and public involvement, inconsistent sensor quality and uncontrolled sampling conditions caused variations in accuracy. **[3]**

Hu et al. created a fine-grained sensing network for smart cities, showing strong scalability and wide coverage. Despite this, the system depended on expensive sensor nodes and centralized cloud servers, limiting its adoption for personal or low-budget use. **[4]**

Bulot et al. conducted laboratory tests on low-cost particulate matter sensors and found that they can capture pollution spikes effectively. Over time, however, accuracy declined due to humidity and particle composition issues, highlighting the need for calibration. **[5]**

Alfano et al. carried out a large review of low-cost air quality sensors and support the idea that affordable sensors require frequent calibration. They also noted that environmental changes, such as temperature and humidity, significantly affect readings. **[6]**

González et al. developed a LoRa-based wireless network for city-wide monitoring, capable of covering large distances. But the system’s size, infrastructure, and complexity made it inconvenient for individual, personal monitoring applications. **[7]**

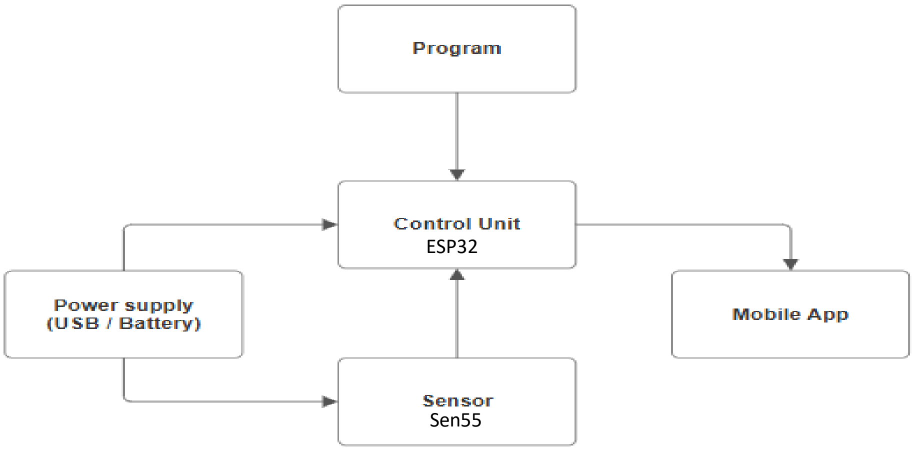
Alvear-Puertas et al. designed a vehicle-mounted portable air quality device that improved mobility and coverage. Even though it was more flexible, it still needed continuous power from the vehicle and lacked detailed particulate matter classification. **[8]**

Cho et al. worked on reducing sensor noise and improving air quality readings using advanced filtering methods. Their system did improve accuracy but was mainly suited for fixed locations rather than handheld or portable devices. **[9]**

Buelvas emphasized the importance of data quality and reliability in IoT-based environmental systems. The study highlighted how poor calibration practices and inconsistent sensor behaviour reduce trust in long-term air quality data from low-cost sensors. **[10]**

PROPOSED PROJECT

The proposed device thus bridges the gap between industrial monitoring systems and personal environmental awareness tools by offering precision, convenience, and accessibility, encouraging healthier lifestyle choices and greater public engagement in environmental protection.



The proposed system focuses on creating a compact and efficient air quality monitoring device that continuously tracks and analyzes environmental conditions in real time. It measures key air quality parameters, calculates the Air Quality Index (AQI), and presents the results through a dedicated mobile application. This setup ensures users receive instant updates and insights on surrounding air quality, promoting awareness of environmental changes and their effects on health. Designed for accuracy and convenience, the project aims to provide a practical and accessible solution for everyday air quality monitoring.

FEASIBILITY STUDY

The feasibility study evaluates the practicality and effectiveness of the proposed air quality monitoring system. The analysis confirms that the project is operationally viable, as the system can function reliably in real-time conditions with minimal maintenance. Economically, the components are affordable and easily available in the market, making the project cost-effective and suitable for large-scale implementation. From a technical standpoint, the design is straightforward and compatible with commonly used hardware and mobile platforms, ensuring easy deployment and user accessibility. Overall, the system is feasible for development, implementation, and real-world application.

METHODOLOGY

The methodology for this project involves a structured approach to designing, developing, and testing an IoT-enabled air quality monitoring system capable of delivering accurate, real-time environmental data. The process begins with component selection, where the ESP32-WROOM-32 microcontroller and the SEN55 environmental sensor are chosen for their performance, reliability, and suitability for continuous monitoring. After selecting the hardware, the next step is system integration, which involves connecting the SEN55 sensor to the ESP32 and ensuring stable digital communication for the transfer of particulate matter, VOC, temperature, and humidity readings.

Once the hardware connections are established, the firmware development phase begins. In this stage, the ESP32 is programmed to collect sensor data at regular intervals, filter and validate the readings, and apply Air Quality Index (AQI) calculations. Efficient data-handling techniques and calibration routines are implemented to ensure accuracy. Following this, the device is equipped with IoT functionality by developing code for wireless data transmission using the ESP32's built-in Wi-Fi capabilities.

The next phase focuses on creating the mobile application interface that receives the processed data from the device. The app is designed to display live air quality metrics, graphs, and alerts in a clear and user-friendly format. Lightweight communication protocols are used to ensure fast updates and low latency between the hardware and the app.

After the hardware, firmware, and application development are complete, the assembled system undergoes testing and validation. This includes checking sensor accuracy, assessing AQI calculations, verifying wireless transmission reliability, and ensuring stable long-term operation in both indoor and outdoor environments. Any errors or inconsistencies are corrected through iterative refinement.

Finally, the system is evaluated for portability, energy efficiency, and usability. Adjustments are made to the enclosure design, power management, and interface layout to ensure the device is practical for everyday use. Through this step-by-step development process, the methodology ensures that the project results in a functional, reliable, and scalable air quality monitoring system capable of providing meaningful insights into environmental health.

Weekly Report

AUG 1–15: Topic Selection & Basic Understanding

- Explored different project ideas and decided to work on air quality monitoring.
- Researched why air quality has become a major issue globally and in India.
- Studied basic concepts such as AQI, particulate matter, VOCs, and their health effects.
- Read general articles and government websites to understand pollution trends.

AUG 16–31: Background Study & Existing Solutions

- Researched existing air monitoring systems used by government and private companies.
- Compared features, limitations, and gaps in current products/apps.
- Started exploring IoT-based air monitoring systems at a high level.
- Collected initial points for synopsis introduction and problem statement.

SEP 1–15: Literature Review & Paper Collection

- Gathered research papers related to IoT-based air quality monitoring.
- Identified common methods used in studies and their limitations.
- Learned the basics of AQI standards (Indian AQI, US AQI).
- Made notes for literature review and objective formulation.

SEP 16–30: Component Exploration (Basic Only) & Synopsis Drafting

- Looked at possible sensors (SEN55, PMS5003, SDS011) just to understand options.
- Explored ESP32 and microcontroller basics without deep technical work.
- Created the structure for the synopsis: Abstract, Introduction, Objectives, Methodology.
- Drafted the first version of methodology (high level and theoretical).

OCT 1–15: Writing & Refining the Synopsis

- Wrote detailed sections: Introduction, Literature Review, Problem Statement.
- Added points on environmental impact, project need, and scope.
- Refined conceptual methodology — how the system *will* work in theory.
- Corrected structure, added keywords, and improved flow.

OCT 16–31: Final Editing & Completion

- Completed remaining sections: Expected outcomes, Project scope, Timeline.
- Did final proofreading, formatting, and corrections.
- Prepared the final version of the synopsis for submission.
- Ensured all research content was clear, simple, and non-technical as required.

HARDWARE REQUIREMENTS

The following is our hardware requirements:

Name	Details
Microcontroller	ESP32-WROOM-32
Sensor	SEN55 Environmental Sensor
Power Supply	5V Supply (USB or adapter)
Display	Web Interface (Software-based)
Connectivity	Wi-Fi (ESP32)

SOFTWARE REQUIREMENTS

The following is our software requirements:

Name	Details
Operating System	Windows
Platform	Android Application
Development Environment	ESP IDF
Language	Embedded C, Python

FEATURES

• Provides real-time data on air quality, helping users stay informed about environmental conditions.
• Compact and cost-effective solution.
• User-friendly web interface for easy monitoring of AQI.

• Enables timely interventions and protective measures during periods of poor air quality.

• Supports data-driven decision-making for environmental agencies and policymakers.

• Contributes to public health by raising awareness and promoting cleaner air initiatives.

COST ESTIMATION

Cost of entire project is between ₹5000-₹7000

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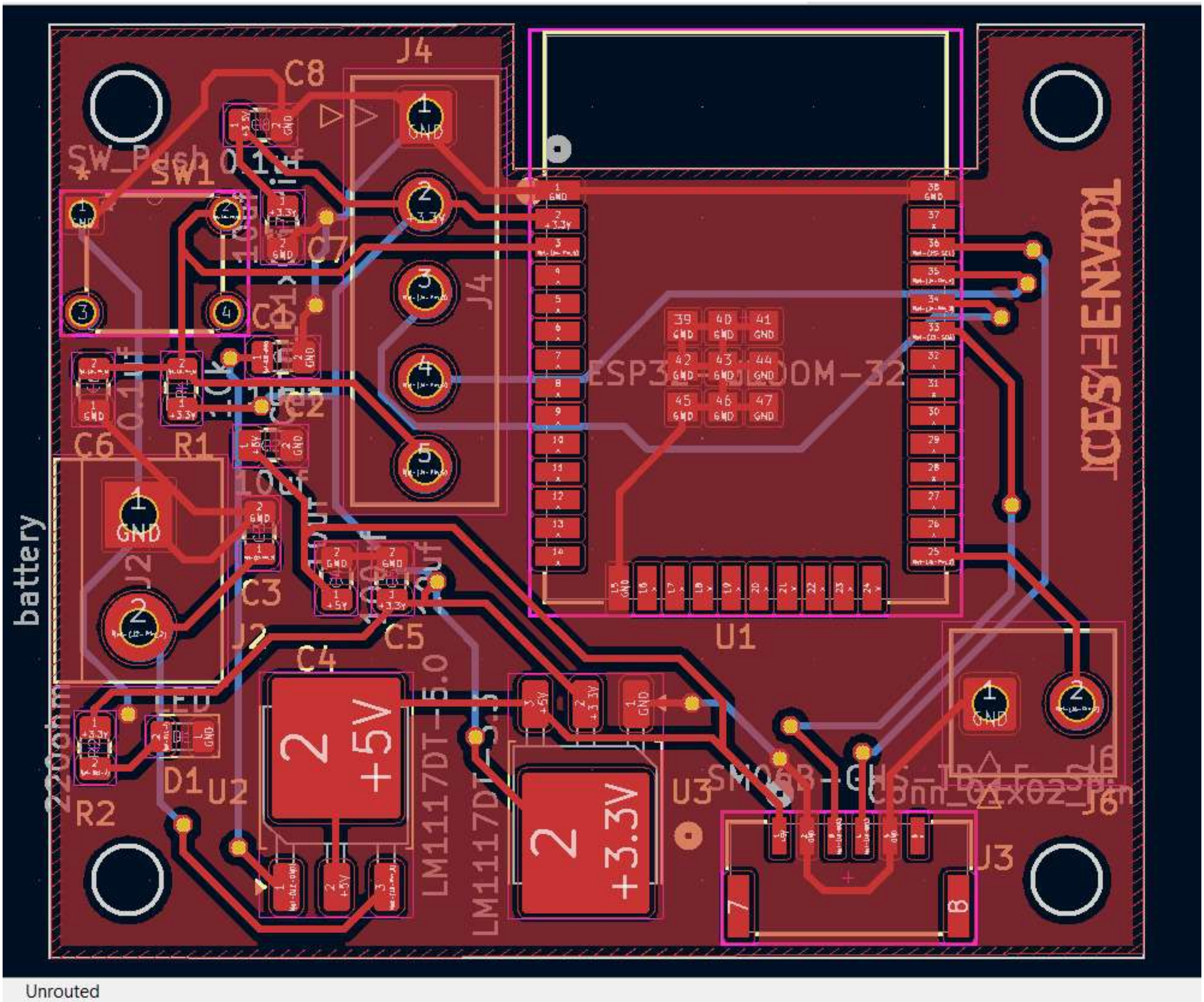
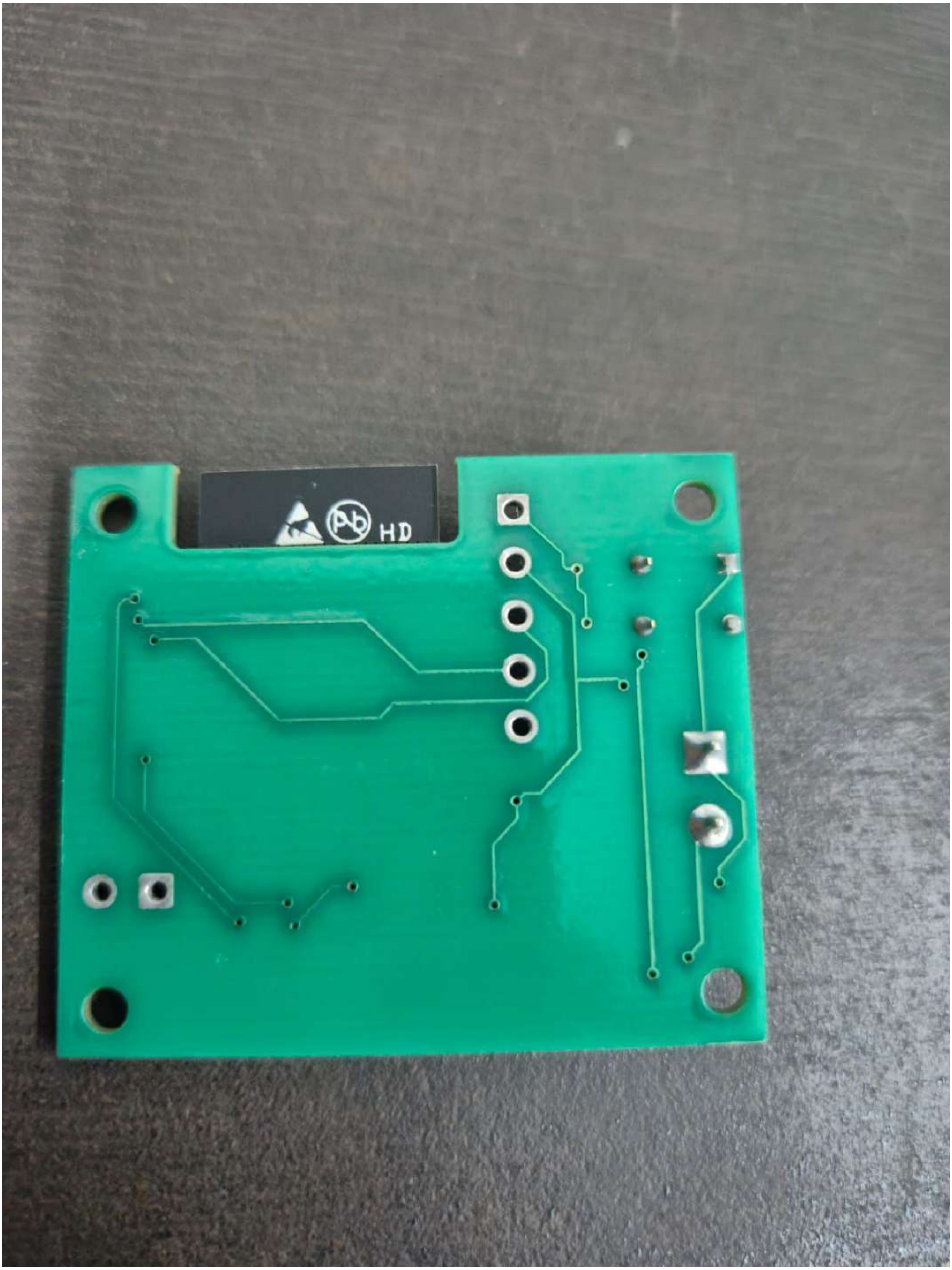


J4
C8
SW_Push
SW1
0.1uf
C7
C6
R1
C2
C3
C5
C4
battery
J2
U1
ESP32-WROOM-32
ICES-ENV01
D1
U2
U3
R2
220ohm
LM1117DT-5.0
LM1117DT-3.3
SM06B-GHS-TB
Conn_01x02_Pin
J6
J3
+5V
+3.3V
GND
Unrouted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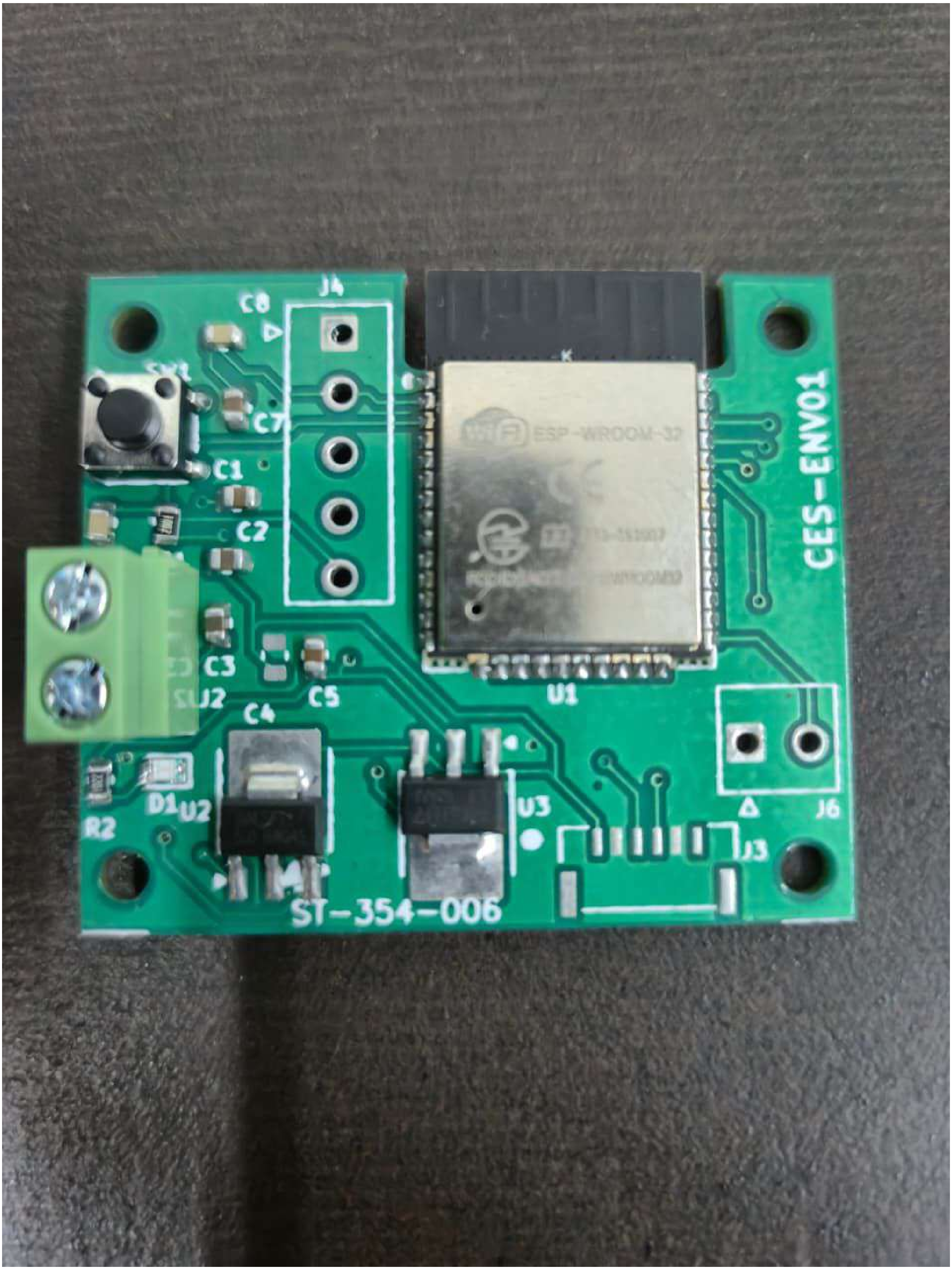
J4
C8
SW1
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C2
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J2
C4
C5
U1
ESP-WROOM-32
CES-ENV01
J6
U3
J3
D1
U2
R2
ST-354-006


AQI INDEX
34
Good
Updated 5 mins ago
DETAILED METRICS
PM1.0
5.0 µg/m³
PM2.5
8.4 µg/m³
PM4.0
12.1 µg/m³
PM10
18.5 µg/m³
VOC
142 Index
NOX
12 Index
HUMIDITY
48 %
TEMP
22.5 °C

A compact PCB based solution for accurately sensing and monitoring air quality in real time.

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Abstract - Air quality helps maintain health conditions for the public; however, the growing trend of industrialization, urbanization, and the increasing use of vehicles have led to a decrease in the quality of the atmosphere over the past years. Breathing polluted air for an extended period can lead to health conditions such as respiratory health issues, heart conditions, allergies, and increased environmental hazards such as the development of smog with an imbalance in the climate. Despite these health hazards, very few people know the actual status of the atmosphere because very limited monitoring devices are available that can monitor the atmosphere with low cost. The aim of this research proposal is to design an efficient low cost atmosphere monitoring device that can monitor critical factors such as the mass of particles with an aerodynamic diameter of less than ten micrometers, carbon monoxide, carbon dioxide, temperature, and humidity. The designed device gives real time information with a simple interface so that the individual can take proper action such as increasing ventilation systems inside the house or not going out much when the atmosphere turns bad. The designed device can be applied both inside the house, colleges, offices, health organizations, and public areas.

Index Terms - Air quality monitoring, embedded systems, environmental sensing, internet of things.

INTRODUCTION

Air pollution, with all the rapid urbanization, growth in industries, and ever increasing vehicular traffic, has become a major global health hazard and threat to environmental stability. Long term exposure to polluted air contributes to respiratory and cardiovascular diseases, besides other chronic diseases. The fine particulate matter and the harmful gaseous pollutants pose a severe threat to human lives since they are capable of piercing through the respiratory system, thus creating a dire need for continuous monitoring of air quality for risk assessment and preventive action.

Recent research has investigated air quality monitoring using Internet of Things based sensing systems. Macro scale solutions with both mobile and fixed sensors enhance spatial coverage and real time data availability [1], whereas Low Power Wide Area Network based systems enable environmental monitoring over a wide area [2]. Participatory and smart city oriented platforms further facilitate fine grained particulate matter sensing across urban environments [3], [4]. However, most of these systems rely on a number of complicated infrastructures, high costs, or centralized architectures that make them unsuitable for personal and localized monitoring.

Indoor and localized monitoring systems, standards driven architectures have been accordingly proposed to address such limitations [5], [6]. Nevertheless, compact, affordable, and easily deployable solutions that emphasize reliable hardware integration and real time accessibility are still in demand.

In this work, a compact, PCB based IoT enabled air quality monitoring system was designed to measure key environmental parameters and provide real time information through a user friendly interface.

The remainder of this paper is organized as follows. Section II reviews related work, Section III describes the system architecture and hardware design, Section IV presents implementation details and results, and Section V concludes the paper with future directions.

RELATED WORK

Air quality monitoring has been extensively explored using Internet of Things architectures that utilize various platforms, processing, and communication components. A real time air quality monitoring and prediction system using both mobile and fixed node sensors was proposed by Zhang and Woo in [1]. Their design combines various air quality sensors with embedded processing, along with cloud resources, for citywide coverage and air quality predictions. Although such distributed designs work very effectively for air quality monitoring at the city level, these designs require various air quality sensors, network communication, and central processing, making them complex. These aspects, therefore, imply the necessity for stand alone air quality monitor designs at the local level.

Based on a similar context, a low power wide area network based air quality measurement system was developed using a distributed sensor network supported by LPWAN technology by Zheng et al. [2]. In this study, an embedded controller was used along with gas and particulate sensors to measure air quality and facilitate long range communication at a lower power consumption cost. However, it was found to be inefficient when considering portable air quality measurement setups valuing lower latencies.

A participatory framework for PM2.5 sensing in smart cities was proposed in [3], where sensing units deployed by citizens measure particulate matter. It is based on varied hardware platforms. In participatory sensing, resolution is improved, but issues of varied hardware deployment or sensing reliability are raised. These issues underscore the significance of controlled hardware integration.

Hu et al. [4] designed a fine grained air quality monitoring system for smart cities by using several nodes, gateways, and server systems. The proposed work mainly targets designing a scalable network and optimizing data in the urban environment. However, such systems include complicated coordination of systems and a higher cost of deployment, making them less suitable for individual monitoring systems for low cost and home usage.

For an indoor setup, a multi point indoor air quality measurement system based on Internet of Things was presented by Liu et al. in [5]. In this system, multiple embedded controllers are used, which are then linked to air quality sensors in multiple rooms. While this greatly helps in analyzing an indoor setup, it also leads to a need for simpler designs involving multiple functionalities in a single component.

Kumar et al. [6] presented an air quality monitoring system based on ISO/IEC/IEEE 21451 standards, emphasizing standardized sensor interfaces and interoperable system design. While standards based architectures improve modularity and compatibility, they often result in more complex hardware configurations, underscoring the relevance of simplified PCB level integration for practical deployments.

In contrast to large scale networks, multi node architectures, and standards heavy implementations, the present work focuses on the design of a compact, PCB based air quality monitoring system that integrates sensing, processing, and communication within a single portable device. This design approach targets localized monitoring scenarios where simplicity, deployability, and hardware integration are primary considerations.

PROPOSED SOLUTION

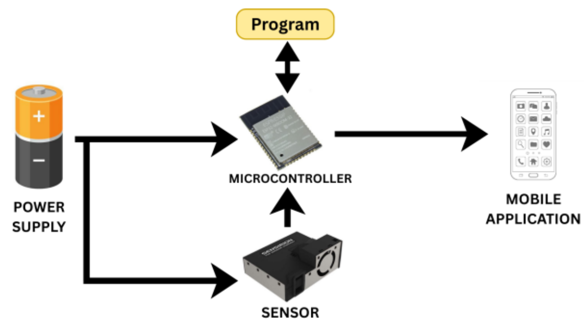

FIGURE 1. THE STRUCTURE OF THE PROPOSED AIR QUALITY MONITORING SYSTEM.

The proposed system would be a compact Internet of Things enabled air quality monitoring device, incorporated with a focus on PCB level integration and real time accessibility. It would integrate sensing, processing, and wireless communication capabilities into a single device, allowing the monitoring of air quality at a local scale for both indoor and near outdoor settings.

The central part of the system comprises an embedded micro-controller, which deals with communication from sensors, data acquisition, as well as networking. An environmental sensor module forms one of the essential parts of the system. This module communicates effectively with the micro-controller to determine the major aspects of environmental air quality. These aspects entail particulate matter concentration, gases, temperatures, and humidity. The system design involved a custom-made PCB. This PCB involves an easy and effective means of power supply delivery and signal transfer from one component to the other. The system also includes wireless communication, which facilitates the transfer of data from the parameters calculated to a user interface. The interface allows users an easy means through which they can access air quality aspects. By combining sensing, processing, and communication aspects through the use of a PCB, the proposed system intends to be easy to use. The system can be applied effectively in the residential, office, educational, hospital, and other indoor setups. These setups require localized information regarding air.

IMPLEMENTATION

This section highlights the implementation of the proposed air quality monitor system in relation to hardware and software elements. There are two primary parts involved in the implementation process. To begin with, the hardware implementation of the proposed system is explained in detail with regards to sensor circuit integration, microcontroller circuit integration, power handling circuit integration, and wireless communication interfaces. Following that is the description of software implementation that entails firmware design development in the proposed system as well as mobile application design development.

I. Hardware design

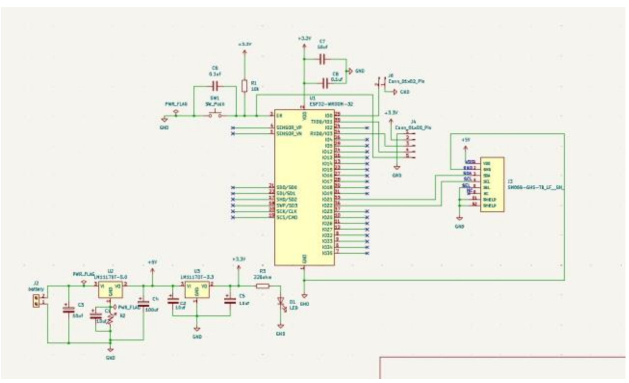

FIGURE 2. THE SCHEMATIC DIAGRAM OF THE PROPOSED AIR QUALITY MONITORING SYSTEM.

The subsystems of the air quality monitoring modules are presented in Fig. 2, and the functions of the sensors are described as follows:

Environmental Sensor (SEN55)

- Quantifies the concentration of particulate matter in PM1.0, PM2.5, PM
- Particle matter levels range of measurement 0-1000
- Offers temperature and humidity readings
- Measuring temperature range -10°C to +60°C, typical accuracy of ±0.5°C
- Relative Humidity Measurement Range: 0 – 90% RH, Accuracy ±3% RH
- Creates volatile organic compounds index and nitrogen oxides index through internal processing
- Facilitates digital communication with the microcontroller

Microcontroller (ESP32 WROOM-32)

- Acts as the central processing unit of the system
- Dual core processor operating at up to 240 MHz
- Supports Wi-Fi (IEEE 802.11 b/g/n) and Bluetooth (BR/EDR and BLE)
- Operates within a supply voltage range of 3.0–3.6 V
- Handles sensor interfacing, data processing, and wireless transmission

Power Supply Unit

- Provides regulated low voltage power to system components
- Ensures stable operation of the microcontroller and sensor module
- Power regulation and distribution are implemented on the PCB

TABLE 1. SEN55 TO ESP32 PIN MAPPING.

S.No.	SEN 55	Pin Name	Description	ESP32 Connection	SEN55 Pin No.
1.	Pin 1	VDD	Supply voltage (5V ±10%)	5V / VIN pin	Pin 1
2.	Pin 2	GND	Ground	GND	Pin 2
3.	Pin 3	SDA	Serial data input/output (I²C)	GPIO 21 (SDA)	Pin 3
4.	Pin 4	SCL	Serial clock input (I²C)	GPIO 22 (SCL)	Pin 4
5.	Pin 5	SEL	Interface select	GND	Pin 5
6.	Pin 6	NC	Do not connect	Not connected	Pin 6

TABLE 2. HARDWARE REQUIREMENTS.

S.No.	**Name**	**Details**
1.	Microcontroller	ESP32-WROOM-32
2.	Sensor	SEN55 Environmental Sensor
3.	Power Supply	5V Supply (USB or adapter)
4.	Display	App Interface (Software based)
5.	Connectivity	Wi-Fi (ESP32)

II. Software development

The software development of the proposed air quality monitoring system is described as follows:

- Firmware Logic and AQI Computation

The ESP32 firmware interacts with the SEN55 sensor to read the values of PM1.0, PM2.5, PM4.0, PM10, temperature, humidity, and VOC Index by using the digital interface.

The firmware calculates particulate matter values and the Air Quality Index by using the linear interpolation formula:

$$AQI = ((I_2 - I_1)/(B_2 - B_1)) * (P - B_1) + I_1.$$

where P represents the measured concentration of pollutants, and B_1 and B_2 along with I_1 and I_2 represents the lower and upper concentration breakpoints, along with index breakpoints, respectively. The prevalent pollutant sub index is chosen as the final result of the AQI reading. Computations are performed in the main firmware loop.

- Wi-Fi Communication
  The ESP32 employs its Wi-Fi module for transmitting data from sensors and calculated AQI values to mobile applications. The data packets are transmitted in fixed time intervals through a lightweight communication protocol to ensure low latency and stable data transfer.

- Mobile Application
  Data received from ESP32 is displayed on this application with real time PM values, environmental conditions, and AQI. It is an application that behaves solely as an interface with no data processing functions.

TABLE 3. SOFTWARE REQUIREMENTS.

S.No.	Name	Details
1.	Operating System	Windows
2.	Platform	Android Application
3.	Development Environment	ESP IDF
4.	Language	Embedded C, Python
5.	Operating System	Windows

CONCLUSION

In this paper, an Internet of Things-based air quality monitoring system has been introduced that is compact and suited for localized environments. Through the integration of environmental observation, processing, and wireless communication on a PCB, an integrated system that provides real time air quality measurement and reporting is viable. Through such an integrated system, it has been made evident that it is feasible and useful for unceasing particulate matter, as well as environmental, observation. The system introduced in the article can be used efficiently for indoor, as well as near outdoor, environments such as residences, offices, and educational institutions, facilitating timely awareness about air quality environments.

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